

Macro I

NAME:

Problem Set 5

Submit Q1-Q5; TA session will cover Q6-Q7; Extra practice Q8

Question 1: Labor market basics using a numerical example

Assume that workers preferences can be described using the utility function

$$u(C, N) = \frac{C^{1-\sigma}}{1-\sigma} - \frac{1}{\psi} N^\psi$$

where C is consumption, N is the labor input, and $\sigma > 0$ ('sigma') and $\psi > 1$ ('psi') are parameters.

The budget constraint is $C = wN$, where w is the real wage in the economy.

In question 5 (for TA session), you are asked to show that the optimal solution implies a labor supply function (for the analysis here you can assume that this is correct)

$$N = w^{\frac{1-\sigma}{\psi-1+\sigma}}$$

Firms producing a Cobb-Douglas production function $Y = AK^\alpha N^{1-\alpha}$ $0 < \alpha < 1$. In question 5 (for TA session), you are asked to show that the optimal labor demand function is (for the analysis here you can assume that this is correct)

$$N = (1 - \alpha)^{\frac{1}{\alpha}} A^{\frac{1}{\alpha}} K \times w^{-\frac{1}{\alpha}}$$

For the purpose of this question, assume: $\psi = 1.2$, $\sigma = 0.7$, $\alpha = 0.3$, $A = 0.4$ and $K = 4$.

1. Solve for the equilibrium levels of w and N (in order to do this you will have to use the equilibrium market clearing condition so that supply equals demand. This should result in one equation where the only unknown is w . Then you can substitute the equilibrium wage in the demand or the supply function to solve for N)
2. Calculate GDP for the economy.
3. What are the new levels of w , N , and GDP if productivity increases by 25% to $A = 0.5$? Describe this equilibrium graphically (i.e. which curve has shifted and why).

4. What are the new levels of w , N , and GDP if A remains at its initial level ($A = 0.4$) and the capital stock increases by 25% to $K = 5$? Describe this equilibrium graphically (i.e. which curve has shifted and why).

Question 2: Foreign aid (of a particular kind)

Suppose that an economy's labor market is in equilibrium, and starts to receive a permanent amount of foreign aid. You may assume that the foreign aid is: (i) in final consumption goods; (ii) delivered directly to households; (iii) expected to last forever (e.g. a positive amount every period).

1. In the short run (i.e. before capital can be adjusted): does the labor supply curve shift? does the labor demand curve shift? Explain why (and in which direction) or why not.
2. What are the short run effects on the real wage, the labor input, and output (i.e. production) in this economy?
3. Are there any effects on future capital accumulation and (therefore) on current investment? Explain.
4. As capital adjusts, what are the implications with regards to the labor market? Specifically, how would the long run equilibrium compare to the short run equilibrium? What about output?
5. In the economy better-off or worse-off as a result of the increase in foreign aid?

Question 3: Income effect on labor supply

In class we described an example for how a temporary positive/negative shock to Total Factor Productivity (TFP) can generate an expansion/recession using the equilibrium model of the labor market and the production function. Here we would like to consider a related example, that is arguably more realistic.

Suppose that the labor market is in an equilibrium. As a result of a new technological progress, TFP is now permanently higher – it “jumps” to a new level and expected to stay at this level forever.

1. Does the labor demand curve shift (in the current period)? If so, in which direction? Explain.
2. Does the labor supply curve shift (in the current period)? If so, in which direction? Explain.
3. As a result of your previous answer, can you tell what will happen to the labor input, the real wage, and GDP?
4. Now suppose that instead of a permanent shock, TFP will be higher for a few periods but will gradually revert back to its initial level. Does this change your answers?
5. Finally, suppose that household preferences are such that there is no income effect on labor supply. Does this change your answers?

6. [NOT FOR SUBMISSION; WILL BE DISCUSSED IN TA SESSION]: show that if the utility function is $U(C, N) = \ln \left(C - \frac{N^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}} \right)$ then there is no income effect on labor supply

Question 4

Assume that the employment and unemployment stocks in the economy behave according to the descriptive model we discussed in class. Denote the job separation/destruction probability by d and the job finding probability by f .

Assume that there are two types of workers in the economy and that all workers are in the labor force (i.e. they are either employed or unemployed):

- 15,000 workers of type A, of which 1,500 are unemployed, and $d_A = 0.07$
- 15,000 workers of type B, of which 1,800 are unemployed.
- the job finding rate is the same across groups: $f_A = f_B$

Use the steady state version of the model to find f (for both types) and d_B .

Question 5 – Task-based production

Suppose that production in the economy can be described as a collection of tasks, and that the production of an individual task j can be described as

$$y(j) = \psi_N(j)N_j + \psi_K(j)K_j$$

where N_j denotes labor employed in task j , K_j denotes capital employed in task j , $\psi_N(j)$ is labor productivity in task j , $\psi_K(j)$ is capital productivity in task j .

In addition, suppose that w and r represent the wage and the rental price of capital, respectively. Both are taken as given by firms.

Suppose that overall output is the sum of four tasks, and that the following represent labor and capital productivity in each task:

Task 1: $\psi_N(1) = 10$; $\psi_K(1) = 2$.

Task 2: $\psi_N(2) = 8$; $\psi_K(2) = 4$.

Task 3: $\psi_N(3) = 5$; $\psi_K(3) = 6$.

Task 4: $\psi_N(4) = 4$; $\psi_K(4) = 8$.

1. Suppose that $w = 100$ and $r = 100$. Which tasks are produced by labor and which ones are produced by capital?
2. Suppose that due to a technological improvement $r = 80$. Is there any change in the division of tasks between capital and labor? Are workers better off/worse off/indifferent?

Question 6 - for TA session - Derive optimal supply and demand functions

The purpose of this question, among other things, is to illustrate how a model of supply and

demand work from a more technical aspect (e.g., when we derive a supply/demand function, what is it a function of? how is it related to optimality? how do we clear the market?)

Assume that workers preferences can be described using the utility function

$$u(C, N) = \frac{C^{1-\sigma}}{1-\sigma} - \frac{1}{\psi} N^\psi$$

where C is consumption, N is the labor input, and $\sigma > 0$ ('sigma') and $\psi > 1$ ('psi') are positive parameters.

The budget constraint is $C = wN$, where w is the real wage in the economy.

Firms producing a Cobb-Douglas production function $Y = AK^\alpha N^{1-\alpha}$, where $0 < \alpha < 1$.

1. Show that the utility function we assumed satisfy the assumptions we made on workers preferences: (i) more consumption is better; (ii) marginal utility from consumption is diminishing; (iii) more labor is worse; (iv) the marginal utility from labor becomes more negative as we work more (i.e it is more 'painful' to work)
2. Assume that firms in the economy are price takers and set labor demand to max profits. Derive the labor demand curve as a function of A , K , α , and w
3. Show that demand increases if A or K increase, and decrease if w increases. Explain why the first two will cause a shift along the curve, and the change in w will be a movement along the curve.
4. Solve for the optimal supply function. For this you will need to solve the worker's optimization problem or use the optimality condition ("static first order condition") that we discussed in class.
5. Use the demand function and the supply function to solve for the equilibrium level of the real wage as a function of A , K and the parameters (σ, ψ, α) .

Question 7 - For TA session - A different immigration example

In class we analyzed the short and long run effects of immigration. Let's look at a related example. Assume an economy that starts at some labor market equilibrium. The economy experiences an inflow of migration so that there is a permanent population increase. The new immigrants are not arriving "empty handed" but with some knowledge and experience that permanently raises the level of productivity in the economy (A).

1. What is the effect, if any, on labor demand in the short run? Explain
2. Describe the various effects on the labor supply curve. Assume that if there is ambiguity then eventually the supply curve shifts to the right (i.e. more supply)
3. Describe the short run equilibrium in the labor market (i.e. what happens to w and N).

4. Turning to the long run effects - will firms choose to invest more? explain why or why not

Question 8: The price of capital goods, labor, and growth

In recent decades economists claimed that cheaper capital goods (e.g. machines) have contributed to growth. This question uses the model(s) we developed in class to evaluate this claim.

Specifically, assume that (i) an economy starts at some equilibrium in the labor market; (ii) firms' current capital K reflects their optimal decision; and (iii) the price of capital p_k has declined permanently. (To simplify, assume that $p_k = p_k^f$ and both are now lower). (iv) In the labor market, the effects of demand are stronger than the effects of supply.

1. What is the effect of the price change on the optimal choice of future capital (K^f) and current investment (I)?
2. Is there an effect on labor demand (i.e. the labor demand curve) in the short run? in the long run? explain
3. Is there an effect on labor supply (i.e. the labor supply curve) in the long run? explain
4. As a result of your previous answers, what happens to N , w and GDP in the long run?

Solution

1. First, when $p_k = p_k^f$ then the user cost is $\text{user cost} = \frac{(r+\delta)p_k}{1-\tau_k}$. Therefore the lower price of capital unambiguously lowers the user cost. In addition, there is no exogenous shock to MPK^f curve. As a result, the new optimal level of K^f is higher. (This is logical... capital is cheaper). In order to increase the stock of capital, firms have to invest more today, and therefore investment should rise as well.
2. Labor demand is unaffected in the short run because there is no change in the *current* A of K . Labor demand will increase in the long run because there will be more capital in the long run, and due to complementarity in production, the marginal product of labor will be higher for any level of N . This will cause a shift of the labor demand curve to the right.
3. The labor supply curve will shift to the left in the long run. The reason is that workers expect higher income permanently. The strength of this effect depends on the strength of the income/wealth effect on labor supply.
4. In the long run, both N and w will be higher. Note that there is an ambiguity here, which is why assumption (iv) is important. Labor demand is higher and labor supply is lower, which means that w will increase but N may be higher or lower. The assumption we made resolves this ambiguity and will result in more N at the new equilibrium.

Future GDP will increase because both K^f and N^F increase.